MATHSCOPE v0.3.1

Stage 1 Foundation - Browser E2E Gate Progress Checklist

ResearchSession -> Schema -> Revision -> Evidence -> IDE Shell

개발 URL: https://project29770.websitepublisher.ai/v0.3.1.html

현재 페이지: version 3 / hash e599b262

안정판: root v0.2.1.1 version 31 / hash e1b6f09f - unchanged

Tracker: blocked at native browser E2E gate

날짜: 2026-10-07

상태	체크리스트	현재 근거
PASS	구현 완료 + 소스/정적 검증 완료	현재 증거로 PASS 처리 가능
PARTIAL	구현 완료, 실제 브라우저 상호작용 검증 대기	Browser/Computer callable tool이 현재 런타임에 노출되지 않음
TODO	아직 시작하지 않음	Stage 2 이후 또는 다음 단계

Tracking rule: 기능이 존재한다는 이유만으로 완료 처리하지 않는다. 브라우저 상호작용 항목은 실제 E2E 검증 전까지 PARTIAL로 남긴다.

1. Stage 1 현재 결론

Stage 1의 데이터 구조, revision/stale engine, evidence trust guards, local-first persistence, IDE Shell markup/runtime은 라이브 v0.3.1 페이지에 구현되어 있다. Foundation inline module의 JavaScript syntax 검사도 PASS했다.

다만 사용자가 연결한 Browser/Computer 플러그인 참조는 이 채팅에 보이지만, 현재 실행 런타임의 callable tool 목록에는 Browser/Computer action과 browser skill이 노출되지 않는다. 따라서 실제 live-page 클릭/DOM/console/screenshot E2E는 실행할 수 없다.

상태	체크리스트	현재 근거
PASS	Stable root isolation	index.html hash e1b6f09f 유지
PASS	Foundation live source	/v0.3.1.html version 3
PASS	Foundation module syntax	PASS
PASS	ResearchSession/schema/revision/evidence runtime	source presence + validators
PARTIAL	IDE Shell native interaction	브라우저 실제 click/state QA 필요
PARTIAL	WebMCP native discovery/invocation	등록 코드는 존재, compatible browser/agent invocation 필요
TODO	Stage 2 default-on	Stage 1 E2E acceptance PASS 후 시작

2. ResearchSession

상태	체크리스트	현재 근거
PASS	ResearchSession type	createDefaultSession()/schema validator
PASS	id	createDefaultSession()/schema validator
PASS	title	createDefaultSession()/schema validator
PASS	goals[]	createDefaultSession()/schema validator
PASS	objects[]	createDefaultSession()/schema validator
PASS	activeObjectRef	createDefaultSession()/schema validator
PASS	representations[]	createDefaultSession()/schema validator
PASS	operators[]	createDefaultSession()/schema validator
PASS	pdes[]	createDefaultSession()/schema validator
PASS	spectra[]	createDefaultSession()/schema validator
PASS	geometries[]	createDefaultSession()/schema validator
PASS	topologies[]	createDefaultSession()/schema validator
PASS	kclasses[]	createDefaultSession()/schema validator
PASS	claims[]	createDefaultSession()/schema validator
PASS	evidence[]	createDefaultSession()/schema validator
PASS	assumptions[]	createDefaultSession()/schema validator
PASS	revisions[]	createDefaultSession()/schema validator
PASS	checkpoints[]	createDefaultSession()/schema validator

상태	체크리스트	현재 근거
PASS	environmentRef	createDefaultSession()/schema validator

2.1 Session operations

상태	체크리스트	현재 근거
PARTIAL	Create Session	createSession() 구현 / UI 버튼 연결
PARTIAL	Save / Load Session	persist() + startup loadStore()/ensureSession()
PARTIAL	Duplicate / Rename / Archive	각 handler 구현
PASS	Session revision model	createSessionRevision()
PARTIAL	Checkpoint create/restore	snapshot/restore handler
PARTIAL	Autosave + dirty indicator	700ms debounce / DIRTY-CLEAN UI
PARTIAL	Crash recovery	CRASH_KEY snapshot + recovery bar
PARTIAL	Import / Export	Foundation JSON + 5 MiB safety cap

3. Typed Object / Representation contract

3.1 ObjectSpec

상태	체크리스트	현재 근거
PASS	type	ObjectSpec validator 포함
PASS	base field	ObjectSpec validator 포함
PASS	intrinsic dimension	ObjectSpec validator 포함
PASS	ambient dimension	ObjectSpec validator 포함
PASS	structure metadata	ObjectSpec validator 포함
PASS	domain/manifold ref	ObjectSpec validator 포함
PASS	revision id	ObjectSpec validator 포함
PASS	assumptions	ObjectSpec validator 포함
PASS	provenance	ObjectSpec validator 포함
PASS	hash	ObjectSpec validator 포함

3.2 RepresentationSpec

상태	체크리스트	현재 근거
PASS	sourceObjectRef	RepresentationSpec validator 포함
PASS	method	RepresentationSpec validator 포함
PASS	displayDim	RepresentationSpec validator 포함
PASS	mapDefinition	RepresentationSpec validator 포함
PASS	parameters	RepresentationSpec validator 포함
PASS	sampling	RepresentationSpec validator 포함
PASS	injectivityStatus	RepresentationSpec validator 포함
PASS	ambiguityRef	RepresentationSpec validator 포함
PASS	fidelityVector	RepresentationSpec validator 포함
PASS	provenance	RepresentationSpec validator 포함
PASS	revision	RepresentationSpec validator 포함

3.3 Representation methods

상태	체크리스트	현재 근거
PASS	projection	REPRESENTATION_METHODS enum
PASS	slice	REPRESENTATION_METHODS enum
PASS	embedding	REPRESENTATION_METHODS enum
PASS	stereographic	REPRESENTATION_METHODS enum

상태	체크리스트	현재 근거
PASS	PCA-like reduction	REPRESENTATION_METHODS enum
PASS	spectral reduction	REPRESENTATION_METHODS enum
PASS	user-defined map	REPRESENTATION_METHODS enum
PASS	sampled finite representation	REPRESENTATION_METHODS enum

3.4 Representation guards

상태	체크리스트	현재 근거
PARTIAL	3D render requires valid RepresentationSpec	guard 구현, live click 미검증
PARTIAL	Known non-injective projection badge	Foundation canvas warning UI
PASS	Lost dimensions	lostDimensions
PASS	Discarded coordinates	discardedCoordinates / parameters
PASS	Inverse status	inverseStatus
PASS	Ambiguity fibre	ambiguityRef

4. Revision / Dependency / Stale Engine

4.1 Revision record

상태	체크리스트	현재 근거
PASS	revision id	createRevisionRecord()
PASS	parent revision	createRevisionRecord()
PASS	diff/delta	createRevisionRecord()
PASS	timestamp	createRevisionRecord()
PASS	author/system source	createRevisionRecord()
PASS	environment hash	createRevisionRecord()
PASS	invalidated edges	createRevisionRecord()
PASS	superseded edges	createRevisionRecord()

4.2 Stale propagation

상태	체크리스트	현재 근거
PASS	Representation -> STALE	object revision invalidation
PASS	Claim -> REVIEW	dependency change
PASS	Numerical evidence -> STALE	staleEvidence()
PASS	FORMAL PASS -> HISTORICAL FORMAL	과거 revision evidence 보존
PASS	Previous theorem edge preservation	theoremEdgePreserved
PARTIAL	Current revision unsupported warning	UI 실제 E2E 필요

4.3 Dependency regression fixtures

상태	체크리스트	현재 근거
PASS	object change	runDependencyTests() fixture
PASS	assumption change	runDependencyTests() fixture
PASS	representation change	runDependencyTests() fixture
PASS	mesh change	runDependencyTests() fixture
PASS	solver tolerance change	runDependencyTests() fixture
PASS	boundary condition change	runDependencyTests() fixture
PASS	operator domain change	runDependencyTests() fixture
PASS	Lean environment change	runDependencyTests() fixture

5. Evidence / Trust Ladder

상태	체크리스트	현재 근거
PASS	FORMAL PASS	EVIDENCE_GRADES
PASS	THEOREM-BACKED	EVIDENCE_GRADES
PASS	CERTIFIED NUMERICAL	EVIDENCE_GRADES
PASS	NUMERICAL INDICATOR	EVIDENCE_GRADES
PASS	EMPIRICAL CORRESPONDENCE	EVIDENCE_GRADES
PASS	RESEARCH HYPOTHESIS	EVIDENCE_GRADES
PASS	UNKNOWN	EVIDENCE_GRADES
PASS	STALE	EVIDENCE_GRADES

5.1 Forbidden promotions

상태	체크리스트	현재 근거
PASS	AI -> FORMAL PASS blocked	canPromoteEvidence()
PASS	Numerical -> THEOREM-BACKED blocked without theorem	explicit theoremRef required
PASS	Visual -> topology equivalence blocked	topology evidence required
PASS	LSP no-error -> FORMAL PASS blocked	diagnostic != final verifier
PASS	Finite approximation -> exact infinite spectrum blocked	scope guard

6. IDE Shell

6.1 Top bar

상태	체크리스트	현재 근거
PARTIAL	project/session selector	markup + event handler 존재 / native E2E 대기
PARTIAL	current goal	markup + event handler 존재 / native E2E 대기
PARTIAL	Run	markup + event handler 존재 / native E2E 대기
PARTIAL	Verify	markup + event handler 존재 / native E2E 대기
PARTIAL	Stress Test	markup + event handler 존재 / native E2E 대기
PARTIAL	Command Palette	markup + event handler 존재 / native E2E 대기
PARTIAL	evidence summary	markup + event handler 존재 / native E2E 대기
PARTIAL	environment status	markup + event handler 존재 / native E2E 대기

6.2 Object Tree

상태	체크리스트	현재 근거
PARTIAL	mathematical object	typed group shell 존재 / live interaction 대기
PARTIAL	representations	typed group shell 존재 / live interaction 대기
PARTIAL	function spaces	typed group shell 존재 / live interaction 대기
PARTIAL	operators	typed group shell 존재 / live interaction 대기
PARTIAL	PDE	typed group shell 존재 / live interaction 대기
PARTIAL	spectra	typed group shell 존재 / live interaction 대기
PARTIAL	geometry	typed group shell 존재 / live interaction 대기
PARTIAL	topology	typed group shell 존재 / live interaction 대기
PARTIAL	K-theory	typed group shell 존재 / live interaction 대기
PARTIAL	claims	typed group shell 존재 / live interaction 대기
PARTIAL	evidence	typed group shell 존재 / live interaction 대기

상태	체크리스트	현재 근거
TODO	entropy dedicated node	Stage B semantics 전까지 보류
TODO	symmetry dedicated node	Stage C semantics 전까지 보류

6.3 Main Canvas tabs

상태	체크리스트	현재 근거
PARTIAL	Field	tab shell + handler 존재
PARTIAL	2D	tab shell + handler 존재
PARTIAL	3D	tab shell + handler 존재

상태	체크리스트	현재 근거
PARTIAL	Spectrum	tab shell + handler 존재
PARTIAL	Resolvent	tab shell + handler 존재
PARTIAL	Singularity	tab shell + handler 존재
PARTIAL	Geometry	tab shell + handler 존재
PARTIAL	Topology	tab shell + handler 존재
PARTIAL	Group	tab shell + handler 존재
PARTIAL	Index	tab shell + handler 존재
PARTIAL	Research Graph	tab shell + handler 존재

6.4 Inspector

상태	체크리스트	현재 근거
PARTIAL	type	selected node inspector / later domain data extension
PARTIAL	current revision	selected node inspector / later domain data extension
PARTIAL	assumptions	selected node inspector / later domain data extension
PARTIAL	representation method	selected node inspector / later domain data extension
PARTIAL	fidelity	selected node inspector / later domain data extension
PARTIAL	evidence grade	selected node inspector / later domain data extension
PARTIAL	provenance	selected node inspector / later domain data extension
PARTIAL	stale state	selected node inspector / later domain data extension
PARTIAL	numerical errors	selected node inspector / later domain data extension
PARTIAL	theorem assumptions	selected node inspector / later domain data extension

6.5 Bottom Dock

상태	체크리스트	현재 근거
PARTIAL	PDE	dock shell + handler 존재
PARTIAL	Operator	dock shell + handler 존재
PARTIAL	Spectrum	dock shell + handler 존재
PARTIAL	K-Theory	dock shell + handler 존재
PARTIAL	Lean	dock shell + handler 존재
PARTIAL	Claims	dock shell + handler 존재
PARTIAL	Evidence	dock shell + handler 존재
PARTIAL	Console	dock shell + handler 존재
PARTIAL	Dependency Graph	dock shell + handler 존재

7. WebMCP Foundation integration

상태	체크리스트	현재 근거
PASS	mathscope.session.summary registration	read-only active session summary
PASS	mathscope.foundation.self_test registration	schema + dependency + trust guard test
PASS	mathscope.session.create_checkpoint registration	local state mutation
PARTIAL	Native getTools discovery	current runtime lacks Browser tool
PARTIAL	Native tool invocation	current runtime lacks Browser/agent consumer
PARTIAL	Cleanup lifecycle verification	real browser mount/unmount test pending

8. Browser E2E acceptance matrix - 현재 blocker

상태	체크리스트	현재 근거
PARTIAL	Page identity / title	live metadata는 connector로 확인, real browser URL/title 미확인
PARTIAL	Not blank	source markup 확인, actual rendered DOM snapshot 미확인
PARTIAL	No framework/runtime overlay	JS syntax PASS, actual rendered overlay 미확인
PARTIAL	Console health	callable Browser dev logs 부재
PARTIAL	Screenshot evidence	callable Browser/Computer screenshot 부재
PARTIAL	Session CRUD interaction proof	real click/state validation 필요
PARTIAL	Autosave / recovery proof	real localStorage browser lifecycle 필요
PARTIAL	Checkpoint restore proof	real interaction 필요
PARTIAL	Command Palette proof	real keyboard/click 필요
PARTIAL	Canvas/Dock switching	real interaction 필요
PARTIAL	Guarded legacy S3 3D open	WebGL browser execution 필요
PARTIAL	Foundation self-test UI execution	actual browser execution 필요

9. Stage 2 gate

마스터 체크리스트는 Stage 2를 PDE -> Operator -> finite spectrum -> 3D field의 얇은 end-to-end 제품으로 정의한다. 현재 tracking rule에 따라 Foundation의 IDE/browser acceptance를 PASS하기 전 Stage 2를 default-on으로 진행하지 않는다.

상태	체크리스트	현재 근거
TODO	PDESpec vertical slice	Stage 1 E2E PASS 후
TODO	OperatorSpec vertical slice	Stage 1 E2E PASS 후
TODO	Finite SpectrumSpec	Stage 1 E2E PASS 후
TODO	3D field visualization	Stage 1 E2E PASS 후

10. 현재 외부 연결 상태

상태	체크리스트	현재 근거
PASS	Airtable connection	pong; base 'mathlib4 Remote Verification' 확인
PASS	Render connection	workspace 확인; mathscope-cloud-lean service 존재
PASS	WebsitePublisher connection	project 29770 live page read/write 가능
PASS	Webflow plugin invocation	연결 가능하나 현재 Webflow site 0개 - 이 호스팅에는 사용하지 않음
PASS	Exa verification	5-point Dirichlet finite-difference / discrete spectrum 자료 확인

Stage 2 golden vertical slice 후보로는 unit square Dirichlet Poisson problem과 5-point finite difference가 적합하다. 연속 operator와 finite matrix representation을 명시적으로 분리하면서 PDE -> Operator -> finite spectrum -> 3D surface를 한 세션에서 연결할 수 있다.

11. 정확한 blocker와 다음 액션

Blocker: 사용자 메시지에는 @Browser / @Computer가 첨부되어 있지만, 현재 이 채팅 런타임의 callable tools와 skills 목록에는 Browser/Computer tool 또는 browser skill이 실제로 노출되지 않는다. 따라서 브라우저 검증을 했다고 주장할 수 없다.

필요한 다음 환경: Browser/Computer action이 실제 도구로 노출되는 실행 환경에서 /v0.3.1.html을 열고 위 E2E matrix를 수행한다. 모두 PASS하면 Foundation tracker를 DONE으로 바꾸고 Stage 2를 즉시 시작한다.

현재 tracker: Stage 1 = blocked at E2E gate, source implementation approximately 85%. Stage 2 = held by acceptance gate.